

Chapter 6

Digression on Circular Convolution

6.1 Circular Convolution

- **Problem:** in the time domain, **linear convolution** is complicated to perform.

$$y[n] = (x * h)[n] = \underbrace{\sum_{l=0}^L h[l]x[n-l]}_{\text{discrete time}} \quad (6.1)$$

Where:

- h is a **filter**
- x is the **input signal**
- y is the **output signal**

In the frequency domain, convolution becomes a simple **multiplication**!

$$y[n] = x[n] * h[n] \xrightarrow{\text{FFT}} Y[k] = X[k] \cdot H[k] \quad (6.2)$$

So the idea is:

1. Transform the signal (f domain)
2. Equalize it (do stuff)
3. Transform it back (time domain)

Attention

Multiplication holds only for **circular convolution**, not linear.

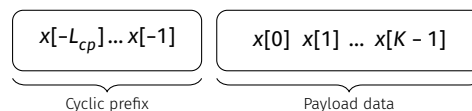
We need a way to “**trick**” the linear convolution channel so that its output corresponds to that of a circular convolution $\Rightarrow$ **CYCLIC PREFIX (CP)**!

Suppose we have a **payload** of length K and a response of the filter h with **memory** or order L .

The **CP** is obtained by taking the last L_{cp} samples of the **payload** and **prepending** them to the start of that block.

6.2 Cyclic Prefix & Fourier Taxonomy

To guarantee that the **linear convolution** is equal to the **circular** one, we must have $L_{cp} \geq L$.



I obtain an **extended signal** $\bar{x}[n]$ of length $K + L_{cp}$. In this way, the **periodic input** necessary for the **circular convolution** is simulated.

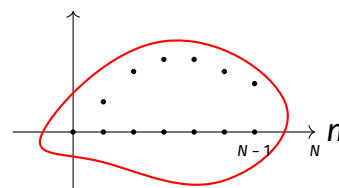
TIME DOMAIN	FOURIER F.T. exists
Periodic $NT = T_p, F = \frac{1}{T_p}$ discrete time T aperiodic	$S(kF) = \sum_{n=0}^{N-1} s(nT)e^{-j2\pi kFnT}$ $S(f) = \sum_{n=-\infty}^{+\infty} s(nT)e^{-j2\pi fnT}$
periodic signals $T_p, F = \frac{1}{T_p}$ continuous time t aperiodic signals	$S(kF) = \int_0^{T_p} s(t)e^{-j2\pi kt} dt$ $S(f) = \int_{-\infty}^{+\infty} s(t)e^{-j2\pi ft} dt$ ↑ this is a generic case

We focus on the behavior at **discrete periodic time**.

discrete time periodic:

$$\sum s(n)e^{-j2\pi \frac{kn}{N}} \quad (6.3)$$

→ Also called **Discrete Fourier Transform (DFT)**.



The **circular convolution** is defined as:

$$c(n) = \sum_{l=0}^{K-1} h[l]x[(n-l)_N] \quad (6.4)$$

Relationship in frequency domain:

$$C(k) = H(k) \cdot X(k) \quad (6.5)$$

6.3 Graphical Representation: SC-FDE

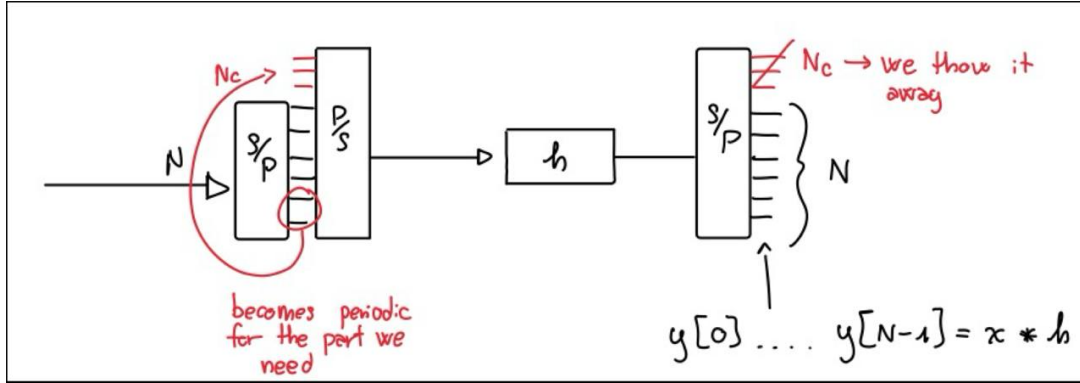


Figure 6.1: SC-FDE

6.3.1 Receiver Structure Notes

- **Simple transmitter**, complicated receiver.
- We can put the equalizer in the transmitter to make it more balanced.

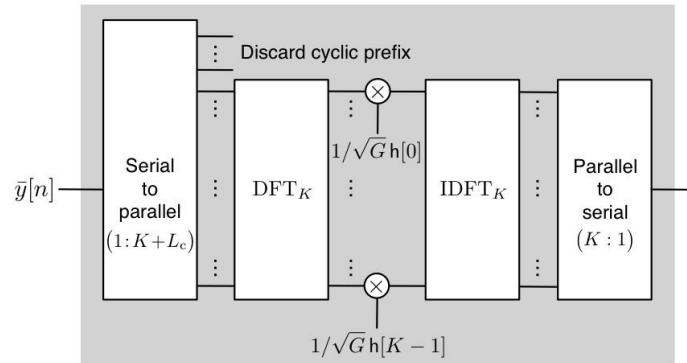


Fig. 2.14 Top: Serial-to-parallel, DFT, Equalization, IDFT

6.3.2 Noise Variance Derivation

$$= \frac{N_0}{G} U^* \text{diag} \left(\frac{1}{|h[0]|} \dots \frac{1}{|h[K-1]|} \right) \text{diag} \left(\frac{1}{h[0]} \dots \frac{1}{h[K-1]} \right)^* U \quad (2.197)$$

$$= \frac{N_0}{G} U^* \text{diag} \left(\frac{1}{|h[0]|^2}, \dots, \frac{1}{|h[K-1]|^2} \right) U. \quad (2.198)$$

The above covariance matrix is, in general, not proportional to the identity, indicating that the noise is correlated. The **variance of the noise** on the n -th equalized symbol is:

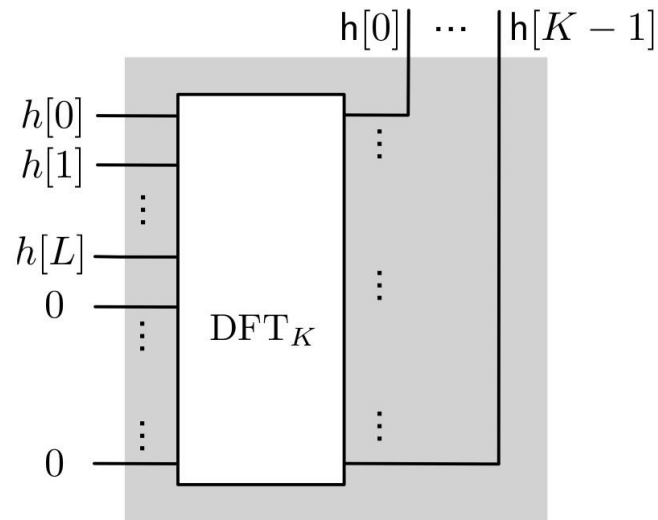


Fig. 2.14 Bottom: Channel taps into DFT

$$\left[\frac{N_0}{G} U^* \text{diag} \left(\frac{1}{|h[0]|^2}, \dots, \frac{1}{|h[K-1]|^2} \right) U \right]_{n,n} \quad (2.199)$$

$$= \frac{N_0}{GK} \sum_{k=0}^{K-1} e^{-j\frac{2\pi}{K}nk} \frac{1}{|h[k]|^2} e^{j\frac{2\pi}{K}nk} \quad (2.200)$$

$$= \frac{N_0}{GK} \sum_{k=0}^{K-1} \frac{1}{|h[k]|^2} \quad (2.201)$$

which does not depend on n , revealing that the noise variance is constant over the output.